

Thematic Analysis Report 3: Technical Summary

This report documents the method, software, AI model, and the exact user prompt used for the analysis.

Method

Raw OCR text files were read from Documents/OCR 1.1/Raw. The analysis ignored author and reference sections, parsed the remaining poster text into sections, and assigned one motivation category per poster using a fixed set of categories defined after reviewing the full poster set. Disease mentions were captured as exact wording from the raw text and then grouped into broader disease-state categories.

Software and model

AI model	GPT-5.4 mini
Python	3.12.1 (main, Jan 8 2024, 04:46:10) [Clang 17.0.6]
reportlab	4.4.10
pandoc	pandoc 2.17.1.1
Source files	27 raw poster text files from Documents/OCR 1.1/Raw

Prompt used

I am part of a team of researchers doing a study on student research posters. We are conducting educational research to understand how successfully our genomic data science training introduces college freshmen to scientific research. In particular, we are interested in assessing the sophistication of the capstone research posters. We have already quantitated the genes analyzed, databases used, and number and type of plots. We're now interested in a thematic analysis of the text.

We have a total of 29 posters which I have extracted text from using OCR. Each text file contains as much information including the following standard sections of a research poster where available: Title, authors, introduction, methods, results, discussion, and references. Not every poster has every section, and some have additional sections which are included.

These posters are placed in Documents > OCR 1.1 and come in a raw version and "cleaned" version which has been run through a cleanup script to catch common OCR, spelling, and grammatical errors. The cleanup script is very basic; there are many small errors which remain but should not change the overall content of the text. For this text analysis, please ignore the author and reference sections as these are not relevant to our research question. I have included only the 27 high quality posters in these folders.

Please conduct a thematic analysis of the raw text files. Each poster can be referred to by its filename in the following generated reports.

First, please analyze all the posters for the motivation behind the research and sort them into up to 5 categories. You may choose the categories. Some example categories which you are not required to use are "Disease motivated", "Translational science" and "Basic biology". Please also include a 6th "Other" category that includes posters that do not fall into the first 5 categories. Please only assign one category per poster. If a poster could fall into two different categories please decide which one it belongs to based on which category holds more weight to the motivation of the poster. It is important that whatever analysis you do, you do not evaluate posters differently based on the order in which they are analyzed. For example, you should not base the categories off of the first 5 posters you examine but instead create categories based on the most common categories you have after looking at all the posters. Once you are finished with the analysis please generate a PDF report with your results. The report should include 1) A table with the total count of posters included in each category, a list of the posters included in each category, a short and easy to understand explanation of what each category means (ex. What defines "Basic Biology" research including examples). 2) For each poster please create a bullet-point list of text from the poster that supports assigning it to the category you chose. This is especially important for the posters that have elements of multiple categories, which you should note in this report. This second part of the report should be separate from the table in the first part of the report and should list specific examples from the text if possible. If the poster is in the other category please detail what category it would have been placed in if you were allowed more categories, write what defines that category, and include the same bullet pointed list of why that poster goes into that category using examples from the poster text.

Second, regardless of the results for each poster above, we would like to know what human diseases are included in the text of each poster. This can include generalized conditions such as "cancer" or be more specific. Please include the same

wording that the poster uses. Create a PDF report that includes 1) A table relating the poster to any human diseases that were listed in the text. Please also include posters that did not list human diseases. 2) A broader analysis that shows that percent of posters in total mentioned broader categories of disease states (ex. "Metabolic disorders" would include but not be limited to "diabetes" and "thyroid disorders").

Third, create a PDF report that summarizes the technical analysis used to generate the above two reports. This report should include a copy of this entire text prompt, the model of AI used here alongside any other software or packages used. Include the version of the AI and any other software.

All 3 reports should be placed in a new directory within the OCR 1.1 directory that is titled with the format "AI_model#_Prompt2". Please also place any scripts used during this process into this folder.

Output files

motivation_thematic_analysis.pdf
disease_thematic_analysis.pdf
technical_report.pdf